Domain of the relation {(0,1),(3,22),(90,34)}is _____ .

A. {1,22,34}
B. {0,1,3}
C. {0,1,3,22,90,34}
D. {0,3,90}

Let P(A) be the power set of a non-empty set A and by using the definition of "subset" relation, ⊆, for all X, Y ∈ P(A), X ⊆ Y ⇔ ∀ x, iff x ∈X then x ∈Y. Then _____.

A. ⊆ is a partial order relation.
B. ⊆ is not a partial order relation.

Let R be the universal relation on a set A then which one of the following statement about R is true?

A. R is not reflexive
B. R is not transitive
C. R is reflexive, symmetric and transitive.
D. R is not symmetric

Let A={1 , 2 , 3 , 4} and R={(1 , 1)(2 , 2)(3 , 3)(4 , 4)} then R is

A. Symmetric
B. Options a,b,and c all true
C. Reflexive
D. Transitive

R={(a,1)(b,2)(c,3)(d,4)} then the inverse of this relation is.....

A. {(a,1)(2,b)(3,c)(4,d)}
B. {(1,a)(2,b)(3,c)(4,d)}
C. {(a,1)(b,2)(3,c)(4,d)}
D. {(1,a)(b,2)(3,c)(4,d)}

Range of the relation {(0,1),(3,22),(90,34)}is _____ .

A. {0,3,90}
B. {0,1,3,22,90,34}
C. {1,22,34}

D. {0,1,3}

Let A = {1,2,3} and R be the relation defined on A such that R ={(1,1), (1,2), (2,3), (3,1)}, then R is _____ .

A. Symmetric

B. Antisymmetric

Let R and S be transitive relations on a set A then _____ .

A. Both R union S and R intersection S are transitive

B. R intersection S is transitive

C. Neither R union S is transitive nor R intersection S is transitive

D. R union S is transitive

There is atleast one loop in the graph of an irreflexive relation.

A. true

B. false

Let R be a binary relation on a set A.R is anti-symmetric iff _____.

A. a, b ∈A if (a,b) ∈R and (b,a) ∈R then a = b

B. a, b ∈A if (a,b) ∈R and (b,a) ∈R then a ≠ b

C. a, b ∉A if (a,b) ∈R and (b,a) ∈R then a ≠ b

D. None of the above

Find x and y given (2x, x + y) = (6, 2).

A. x=2,y=1

B. x=3,y=-1

C. x=2,y=-1

D. x=3,y=1 x=2,y=1

Domain of a relation symbolically written as _____.

A. None of these

B. Dom (R) = {a ∈ A | (a, b) ∈ R}

C. Dom(R) = {b ∈ B | (a, b) ∈ R}

D. Dom (R) = {a ∉ A | (a, b) ∈ R}

Let A = {0,1}and B = {1}.Let R and S be two binary relations on Cartesian product of A and B such that R={(0,1)} and S ={(1,1)}.Then R intersection S = _____ .

A. {1,1}
B. {(0,1)}
C. {0,1}
D. empty

If x ≡ -10 (mod 15) Which of the following integers are valid solution for x ?

A. 7
B. 5
C. 3
D. 6

Let R = {(1, 2), (3, 4), (5, 6), (7, 8)}. Range of the inverse of the relation is _____.

A. {1, 3, 5, 7}
B. {5, 6, 7, 8}
C. {1, 2, 3, 4}
D. {2, 4, 6, 8}

Let R be a relation on a set A. If R is symmetric then its compliment is _____ .

A. Irreflexive
B. Symmetric
C. Reflexive
D. Antisymmetric

If A={1 , 2 , 3} is a set and R = {(1 , 2) , (2 , 2) , (2 , 1)} is a relation on A, R is

A. Reflexive
B. Symmetric
C. None of these
D. Transitive

If a relation R={(1,1)(2,1)(1,2)(2,2)} is given then which of the following is not true about this relation.

A. R is Symmetric.
B. R is Transitive
C. R is Reflexive.
D. R is Irreflexive.

If $x \equiv 17 \pmod 5$ Which of the following integers are valid solution for x ?

A. 4
B. -44
C. 8
D. 12

Let R be a relation on a set A. If R is reflexive then its compliment is _____ .

A. Reflexive
B. Antisymmetric
C. Symmetric
D. Irreflexive

Let the set A = {1,2,3,4} Then the relation {(2,4),(4,2)} is _____.

A. Transitive
B. AntiSymmetric
C. Reflexive
D. Symmetric

The union of the transitive binary relations R = {(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)} and S = {(2,1), (2,4), (3,3)} on the set A = {1,2,3,4} is given by R ∪ S = {(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)}then R ∪ S is_____?

A. not transitive
B. transitive

Let R = {(1, 2), (3, 4), (5, 6), (7, 8)}. Domain of the inverse of the relation is _____.

A. {2, 4, 6, 8}
B. {1, 2, 3, 4}
C. {1, 3, 5, 7}
D. {5, 6, 7, 8}

Let A = {1,2,3,4} and define the relation R on A by R = {(1,2), (2,3), (3,3), (3,4)}.Then _____

A. R is neither reflexive nor irreflexive
B. R is both reflexive and irreflexive
C. R is reflexive
D. R is irreflexive

Range of a relation symbolically written as _____.

A. None of the above
B. Ran (R) = {a ∈ A | (a, b) ∈ R}
C. Ran (R) = {b ∈ B | (a, b) ∈ R}
D. Ran (R) = {b ∉ B | (a, b) ∈ R}

The matrix representation of an inverse relation can be obtained by taking the .....of the relational matrix.

A. transpose
B. Determinant
C. inverse
D. adjoint

Let R be a relation on a set A. R is transitive if and only if for all a,b,c∈A then:

A. (a, b) ∈R and (b, c) ∈R but (a, c) ∉R
B. (a, b)∈R then (b, a) ∉R
C. (a, b) ∈R and (b, c) ∈R then (a, c) ∈R
D. (a, b) ∈R and (b, c) ∈R then (a, c) ∈R(a, b)∈R then (b, a) ∈R

The properties of being symmetric and being anti-symmetric are _____.

A. Negative of each other
B. Not negative of each other

Let A = {1, 2, 3, 4} and define a relation R on A by R = {(1, 1), (1, 2), (1, 3), (2, 3)}.Then which one of the following is a correct statement about R:

A. R is both transitive and reflexive
B. R is neither reflexive nor transitive
C. R is transitive
D. R is reflexive

Let A = {0, 1, 2} and R = {(0,2), (1,1), (2,0)} be a relation on A. Then which of the following statement about R is true?

A. R is reflexive
B. R is symmetric
C. None of the above
D. R is transitive

Which of the following is always true for the matrix representation of a symmetric relation?

A. Matrix has always 0 in its diagonal entries
B. Matrix has always 1 in its diagonal entries
C. Matrix is singular
D. Matrix is equal to its transpose.

If A=(1,2,3) & B=(4,5,6) and R={(1,4)(2,5)(3,6)(3,4)} The complementary relation is .....

A. A × B
B. A × B (intersection) R
C. A × B (difference or -) R
D. A × B (Union) R

Let A = {1,2,3,4} and R = {(1,2), (2,3), (3,3), (3,4)} be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

A. (3,3)
B. (3,4)
C. (2,3)
D. (1,2)

A sequence whose terms alternate in sign is called
Series

Cauchy sequence

Alternating sequence

None of the above

The 3rd term of the sequenceB5=5n

125

625

5

25

If a function is recursively defined as f(0) = 4 & f(n + 1) = 3f (n) + 3 then f(2)=…….

15

55

48

50

The _____ of the terms of a sequence forms a series.

Sum

Multiplication

Let f:R->R is one to one function then c,f, c is not equal 0 is also one to one function.

False

True

720 = ...

4!

5!

7!

6!

The composition of functions is always _____.

Onto

Associative

one-to-one

commutative

What will be the fourth term of the following sequence 1/2 , 2/3 , 3/4 , ___?

5/6

7/8

4/5

6/7

Which of the following set is the domain of a sequence?

Set of natural numbers

Set of complex numbers

Set of real numbers

Set of rational numbers

Let f(X)=x2-1 and g(x)=x+1 define functions f and g from R to R, then (f/g)x

X2-1

X2+1

x-1

x+1

9!/6!=_____.

506

509

504

509

If a function is recursively defined as f(0) = 4& F(n + 1) = 3f (n) + 3 then f(1) =____

25

10

20

15